Game Theory in Electricity Markets

Exercise Sheet 3 — Correlated Equilibria and No-Regret Learning

18th of September 2026

Exercise 1 Coarse Correlated Equilibria and Dominated Actions

Recall that in a Nash equilibrium, no player ever places positive probability on a strictly dominated action. It turns out this is *not* true for coarse correlated equilibria: it is a fact that a CCE can assign strictly positive probability to an action that is strictly dominated. At first glance this seems strange as a CCE is meant to describe a plausible outcome of a learning process, yet it can prescribe an action that is unambiguously bad no matter what the other players do.

Task: Construct a game that has (at least) one strictly dominated action, together with a CCE of that game that assigns strictly positive probability to it. Try to understand why the CCE conditions do not rule this out, even though this can never happen at a Nash equilibrium. Also think about how this may happen within a learning algorithm.

Exercise 2 No-Regret Learning in the Electricity Market

Recall the electricity market bidding game from Exercise 3 of last week's sheet: five generators $G_1, \dots, G_5$ simultaneously submit integer bids $b_i \in \{0, 1, \dots, 80\}$ (EUR/MWh) into the merit-order auction, and generator i's payoff is its profit $\pi_i(b)$.

Task: Implement the multiplicative weights (MW) algorithm and run it *simultaneously* for all five generators (i.e., every generator maintains its own weight vector over its own bid space $\{0, \dots, 80\}$ and updates it every round based on its own payoffs).

(2.1) Before implementing the update rule, first rescale each generator's profit into a loss in $[0, 1]$, e.g. via

$$\ell_i(b) = \frac{\pi_i^{\max} - \pi_i(b)}{\pi_i^{\max} - \pi_i^{\min}}.$$

This rescaling step is *not* optional: the regret bound for MW derived in the lecture only holds when losses lie in $[0, 1]$.

(2.2) Run the algorithm for a suitably large number of rounds T and verify empirically that every generator's average regret converges to zero.

(2.3) Record the empirical distribution of joint bids $(b_1, \dots, b_5)$ over all T rounds, this is a approximate CCE. Compare it to the pure Nash equilibrium (or equilibria) you found last week via best-response: is the CCE's support concentrated on a Nash equilibrium, or spread across other outcomes as well?

Exercise 3 Price of Anarchy of Coarse Correlated Equilibria

Consider the following stylised *curtailment game*, played by $n = 4$ wind farms W_1, W_2, W_3, W_4 during a period of oversupply. Simultaneously and independently, each wind farm i chooses an action $a_i \in \{F, C\}$: either feed its full output into the grid (F), or voluntarily curtail it (C).

Let $k(a) \in \{0, 1, 2, 3, 4\}$ denote the number of wind farms that choose F under the action profile $a = (a_1, \dots, a_4)$. The grid can absorb at most 2 feeding-in wind farms without becoming congested. Payoffs are given by

a_i	condition	$u_i(a)$
C	—	3
F	$k(a) \leq 2$	10
F	$k(a) > 2$	2

(Curtailing earns a fixed compensation of 3; feeding in earns revenue 10 if the grid can absorb it, but only 2 if the grid is congested, since the wind farm must then pay a congestion fee.)

(3.1) Find the social-welfare-maximising action profile(s), i.e. $\arg\max_a \sum_{i=1}^{4} u_i(a)$, and the corresponding optimal welfare OPT.

(3.2) Find all pure Nash equilibria of this game.

(3.3) Formulate the linear program that computes the CCE of *lowest* total welfare.

(3.4) Solve the LP from (3.3) (e.g. numerically, with a solver of your choice) to obtain the worst-case CCE welfare, and report the price of anarchy

$$\text{PoA} = \frac{\text{OPT}}{\min_{\text{CCE}} \text{ welfare}}$$

of this game. Compare it with the pure-Nash-PoA.